Sudipta Chowdhury

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Dhaka, Bangladesh

Research Interests

Augmented Reality (AR), Virtual Reality (VR), Mobile Computing, Resource-Efficient Mobile Systems, and Human-Computer Interaction (HCI).

Experience

- **WellDev Bangladesh Ltd. [🌐]** *April 2024 - Present*
  Software Engineer Dhaka, Bangladesh
  - Developed and maintained native Android and iOS applications for SaaS products serving leading public transportation companies across Switzerland and Germany
  - Implemented MQTT based communication system to replace paper based conventional workflows, enhancing more reliable data exchange
  - Conducted R&D and Knowledge Transfer (KT) sessions to adopt various technologies and practices among the team
  - Mentored and trained interns to learn mobile app development by providing technical guidance
  - Participated in the trainee recruitment process, conducting technical interviews and evaluating candidates' programming and problem-solving skills
  - Designed and developed modular, scalable UI components, business logic, and networking layers, reducing development time by approximately 75–85% through improved code reusability and architectural efficiency.
  - Designed and implemented Jenkins-based CI/CD pipelines with Gitleaks security scanning and integrated Sentry-based logging and monitoring to strengthen software quality, security, and application reliability.
  - Wrote unit tests and maintained over 95% code coverage to support software reliability and maintainability
  - Contributed to business requirement specifications, technical design documentation, and delivery road-maps to support systematic software development
  - Conducted detailed code reviews, providing technical feedback and promoting code quality, maintainability, and adherence to development standards
- **Brain Station 23 [🌐]** *July 2022 - March 2024*
  Associate Software Engineer Dhaka, Bangladesh
  - Developed and maintained native Android applications across fintech and sales domains, contributing to products with over 700k downloads and more than BDT 100 million in monthly revenue
  - Designed and developed reusable UI components and functional screens, optimized application business logic, and integrated backend APIs and Firebase services
  - Implemented fintech features, including bank to mobile wallet transactions, account management, and FAQ modules, alongside merchant discovery functionalities
  - Developed location-aware merchant discovery features by integrating geolocation services, map-based visualization, WebSocket-based bidirectional communication, and backend APIs to support interactive mobile experiences
  - Designed and implemented a sales management dashboard enabling managers to monitor sales performance, inventory levels, pending orders, and operational activities
  - Analyzed existing codebases, diagnosed and resolved software defects, and refactored application components to improve performance, maintainability, and code quality.

Education

- **Chittagong University of Engineering & Technology** *Feb 2018 - July 2023*
  B.Sc in Computer Science & Engineering Chittagong, Bangladesh
  Academic Performance
  - CGPA: 3.24/4.00
  - Final 38 credits CGPA: 3.54/4.00
  - Class rank: 47th out of 126 students (Top 37.3%)

Extracurricular Activities

- Actively participated in competitive programming contests, strengthening algorithmic problem-solving and programming skills
- Contributed to open-source software projects, gaining experience in collaborative development, version control, and community-driven software engineering
- Participated in programming competitions and idea-generation contests, demonstrating technical creativity and innovative problem-solving
- Attended workshops and competitions focused on startup ideation, entrepreneurship, and technology-driven innovation

Research Experience

- **Undergraduate Thesis: Development of an AR-based online shopping system for wearable products**
  Tools: [Unity3D, Google ARCore SDK, AR Foundation, ARCore Augmented Faces API, Android, Python, Hybrid Recommendation System]
  - Developed an AR-based virtual try-on system for wearable products using Unity3D and Google ARCore, leveraging real-time facial tracking and 3D model superimposition to enable users to virtually preview accessories through a smartphone camera
  - Designed and implemented a hybrid product recommendation system integrating content-based and user-based collaborative filtering to generate personalized product suggestions based on product attributes and user purchase histories
  - Applied TF-IDF vectorization and cosine similarity for product-content matching, alongside Pearson correlation and purchase-frequency weighting for identifying similar users and generating recommendations
  - Constructed custom datasets comprising 91 wearable-product 3D models across eyewear, headwear, and jewellery categories, together with 38,949 transaction records to support AR visualization and recommendation experiments
  - Evaluated recommendation performance using Mean Absolute Error (MAE) and Mean Squared Error (MSE), reporting an MAE of approximately 0.63 across 579 users

Teaching Experience

- **Private Tutor (Part-Time)**
  - Provided private tutoring to Higher Secondary Certificate (HSC) students in Mathematics, Physics, and Information and Communication Technology (ICT), supporting their academic learning and examination preparation
  - Taught undergraduate-level computing topics, including Programming Languages, Data Structures, and Algorithms, helping students understand fundamental concepts and develop problem-solving skills

Skills

- **Programming Languages:** C, C++, Kotlin, Java, Swift, Dart, Python, JavaScript, Go
- **Mobile Development:** Native Android, Android SDK, Jetpack Compose, Native iOS, SwiftUI, Flutter
- **Backend Development & APIs:** Node.js, Django, Spring Boot, REST APIs, WebSocket, MQTT
- **Augmented Reality:** Unity3D, Google ARCore SDK, AR Foundation, ARCore Augmented Faces API
- **Databases & Cloud Services:** SQL, Realm, Couchbase, Firebase
- **Testing:** JUnit, Mockito, MockK, XCTest, flutter_test
- **DevOps:** Docker, GitHub Actions, Jenkins, Bitrise
- **Monitoring & Mapping:** Sentry, Mapbox, Geolocation Services
- **Recommendation Systems & Algorithms:** Hybrid Recommendation Systems, Content-Based Filtering, User-Based Collaborative Filtering, TF-IDF, Cosine Similarity, Pearson Correlation
- **Development Tools & Collaboration:** Git, GitHub, Bitbucket, Jira

Achievements and Awards

- **Technical Scholarship** *2019-2023*
  Chittagong University of Engineering & Technology
  - Awarded a technical scholarship in 7 of 8 semesters during undergraduate studies
- **Govt. Board Scholarship** *2010-2015*
  Ministry of Education
  - Awarded a General Grade Government Scholarship in recognition of academic merit in the Primary School Certificate (PSC) Examination

- **Competition Achievement**
  - Received an Honourable Mention and ranked among the top 20 teams out of 120 participating teams in the Student to Startup (2019) national competition, hosted by the ICT Division, Government of Bangladesh
  - Secured 2nd place among 70+ teams at the Mobile App, Game & Job Festival 2023, organized by the ICT Division, Bangladesh
  - Secured 22nd place among 70+ teams at the USTC Inter-University Programming Contest (IUPC) 2020
  - Secured 29th place among 120 teams at the MIST Inter-University Programming Contest (IUPC) 2019
  - Completed the global open-source challenge by contributing 4 validated pull requests to public repositories during Hacktoberfest 2020
- **Competitive Programming**
  - LeetCode Knight — Maximum rating: 1,875; ranked in the top 5-6% [🌐]
  - Codeforces Specialist — Maximum rating: 1,446 [🌐]
  - CodeChef 4-Star Coder — Maximum rating: 1,879 [🌐]

STANDARDIZED TEST SCORES

IELTS Academic: Overall Band 7.0 (Listening: 6.5, Reading: 8.0, Writing: 6.0, Speaking: 7.0)
GRE General Test: 306 (Quantitative: 162, Verbal: 144, Analytical Writing: 3.5)